The Oncogenic Potential of Endometrial Polyps

A Systematic Review and Meta-Analysis

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OBJECTIVE: To systematically review and summarize the medical literature regarding the association of menopausal status, uterine bleeding, and polyp size and risk of malignancy among women undergoing polyp resection.

DATA SOURCES: We supplemented a search of entries in electronic databases with references cited in original studies and review articles to identify studies assessing the risk of malignancy for patients undergoing polypectomy. Key word searches were performed using the words "endometrial polyp," "malignancy," "ultrasound," "saline sonohysterography," "hysteroscopy," and "histopathology."

METHODS OF STUDY SELECTION: We evaluated abstracted data and performed quantitative analyses in observational studies assessing the effects of menopausal status, vaginal bleeding, and polyp size on the risk of malignancy in patients undergoing polyp resection (n=1,552). For each study with binary outcomes, relative risks with 95% confidence intervals (CIs) were calculated. Estimates of relative risk were calculated using fixed and random-effects models. Homogeneity was tested across the studies. Sensitivity analyses were performed to determine the effects of individual studies on the overall effect estimates. Publication bias was assessed using Egger test.

TABULATION, INTEGRATION, AND RESULTS: Seventeen studies met inclusion criteria for this review. Among women found to have endometrial polyps, the prevalence of premalignant or malignant polyps was 5.42% (214 of 3,946) in postmenopausal women compared with 1.7% (68 of 3,997) in reproductive-aged women (relative risk 3.86; 95% CI 2.92–5.11). The prevalence of endometrial neoplasia within polyps in women with symptomatic bleeding was 4.15% (195 of 4,697) compared with 2.16% (85 of 3,941) for those without bleeding (relative risk 1.97; 95% CI 1.24–3.14). Among symptomatic postmenopausal women with endometrial polyps, 4.47% (88 of 1,968) had a malignant polyp in comparison to 1.51% (25 of 1,654) asymptomatic postmenopausal women (relative risk 3.36; 95% CI 1.45–7.80).

CONCLUSION: Based on data from observational studies, both symptomatic vaginal bleeding and postmenopausal status in women with endometrial polyps are associated with an increased risk of endometrial malignancy.
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With the advent of high-resolution transvaginal ultrasonography, saline-infusion ultrasonography, and hysteroscopy, the diagnosis of endometrial polyps has increased.[1–4] Endometrial polyps are detected in premenopausal and postmenopausal women. Some patients are asymptomatic at the time of diagnosis, whereas others present with abnormal bleeding patterns such as intermenstrual bleeding, heavy menstrual bleeding, spotting, discharge, or postmenopausal bleeding.[5,6]

The observation that polyps are rarely diagnosed before menarche and the association between tamoxifen use and endometrial polyps suggest that estrogen stimulation of the endometrium plays an important role in the genesis of endometrial polyps. Several molecular mechanisms have been proposed to play a role in the development of endometrial polyps, including overexpression of endometrial aromatase[7,8] and gene mutations (*HMGIC* and *HMGI[Y]*).[9–13]

A wide array of lesions may present as polypoid growths of the endometrium, the most common being

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the benign endometrial polyp. Other polypoid lesions of the endometrium include endometrial hyperplasia or an endometrial carcinoma that has become polypoid and polypoid adenomyoma.

Once a polyp has been identified, operative hysteroscopy is often the treatment of choice. Some of these surgeries may be avoidable. In asymptomatic premenopausal women, there is a high rate of spontaneous resolution of polyps.[14,15] Similarly, small endometrial polyps less than 1 cm often regress.[15]

In contrast, large endometrial polyps, defined as greater than 1 cm in greatest dimension, have a broader differential diagnosis, including adenofibroma, adenosarcoma, and malignant mixed mesodermal tumor.[16,17]

The majority of polyps are benign, but a small proportion of polyps are malignant. Questions frequently asked by clinicians and patients are, "What is the chance the polyp is cancer?" and "Is removal necessary?"[18] The malignant potential of polyps in relationship to menopausal status and bleeding status has not been adequately addressed by individual published reports.

We conducted a systematic review of research studies evaluating the oncogenic potential of endometrial polyps in premenopausal, postmenopausal, symptomatic, and asymptomatic women. Polyps diagnosed with transvaginal ultrasonography, saline sonohysterography, or hysteroscopy for which histopathologic confirmation of the diagnosis was available were included. Our primary objective was to evaluate the prevalence of endometrioid neoplasia among premenopausal and postmenopausal women found by ultrasonography or hysteroscopy to have endometrial polyps. Our secondary objective was to assess the prevalence of endometrioid neoplasia in women with abnormal bleeding patterns found to have endometrial polyps as well as in asymptomatic women whose polyps were discovered incidentally. We also correlated polyp size with risk of malignancy. The aim of this report, accordingly, is to provide clinicians and women with endometrial polyps with information allowing a more precise estimate of malignancy than is currently available.

SOURCES

We identified observational studies evaluating the risk of malignancy of endometrial polyps in premenopausal, postmenopausal, symptomatic, and asymptomatic women who were identified using computerized databases (PubMed [National Library of Medicine, Bethesda, MD], MEDLINE, and Cochrane Databases) as well as references of published articles

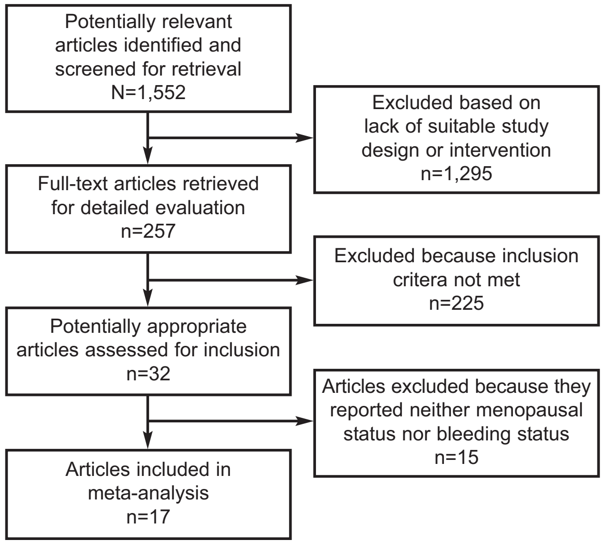

Fig. 1. Flow chart of the study selection.

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and textbook chapters. Searches were conducted for published and unpublished literature between 1980, when transvaginal ultrasonography was introduced into gynecologic practice,[19] and March 2010. There were no language restrictions. Key word searches were performed using the words "endometrial polyp," "malignancy," "ultrasound," "saline sonohysterography," "hysteroscopy," and "histopathology." Figure 1 depicts the flow diagram of studies used in this systematic review. References from studies and review articles were manually screened for relevant citations. Relevant citations were identified. When there were questions regarding the methodology and results of potential articles, the authors were contacted for clarification.

Before performing our systematic review, we used a study protocol and data abstraction form detailing the outcomes to be compared, the subgroups of interest, and the methods and criteria to be used for identifying and selecting relevant studies and extracting and analyzing information. We followed the guidelines for meta-analyses and systematic reviews of observational studies as outlined by the Meta-analysis of Observational Studies in Epidemiology.[20]

STUDY SELECTION

Data were extracted independently from text, tables, and graphs of each of the selected studies. Inclusion criteria included: observational studies of women diagnosed with endometrial polyps by transvaginal ultrasonography, saline infusion sonohysterography, or hysteroscopy who underwent polypectomy with histopathologic diagnosis of their endometrial polyps. Epithelial alterations occurring in endometrial polyps include atrophy, various metaplasias, hyperplasias, and adenocarcinoma (either endometrial or serous papillary carcinoma). The hyperplasias are separated histologically into "simple" or "complex." Each of these is further divided into "atypical" or "without atypia." Simple hyperplasia with atypia is unusual and thus when assessing the endometrium histologically, the gynecologic pathologist is usually considering simple hyperplasia without atypia, complex hyperplasia without atypia, or complex hyperplasia with atypia. Complex hyperplasia with atypia is often difficult to separate histologically from well-differentiated endometrial carcinoma, especially in a curetted or suctioned specimen.

In the articles included in our meta-analysis, endometrial polyps were divided into two groups: benign (endometrial polyps, endometrial polyps with simple hyperplasia [no atypia], endometrial polyps with complex hyperplasia [no atypia]) and premalignant or malignant (endometrial polyps with simple hyperplasia and atypia, endometrial polyps with complex hyperplasia and atypia, and endometrial carcinoma). A recent literature review reported that the frequency of coexistent endometrial cancer among patients with atypical endometrial hyperplasia ranged from 17% to 52%.[21] Given the likelihood of a concurrent carcinoma with atypical endometrial hyperplasia, authors have recommended combining complex atypical hyperplasia with well-differentiated adenocarcinoma into one category (endometrioid neoplasia).[22,23] In this article, the terms "endometrioid neoplasia," "oncogenic," and "malignant and malignancy" refer to polyps with either complex atypical hyperplasia or endometrial carcinoma.

Selected reports assessed the histopathology of endometrial polyps with respect to at least one of the following outcomes: premenopausal women, postmenopausal women, women with abnormal bleeding patterns (including postmenopausal bleeding), or asymptomatic women. Published full-text articles and abstracts were included. Disagreements by authors over inclusion and exclusion of studies and interpretation of data were resolved by consensus reached after discussion, communication, and clarification with two other authors (L.S.R. and A.K.).

Meta-analysis was performed with STATA 8.0 software. The primary outcomes assessed were polyp malignancy among postmenopausal women and women with abnormal uterine bleeding. For each study with binary outcomes, we calculated the relative risk and 95% confidence intervals (CIs) of selected outcomes. Estimates of relative risk for dichotomous outcomes were calculated using fixed-effects (Mantel-Haenszel) and random-effects (DerSimonian and Laird) models.[24,25] The fixed-effects model was used in all outcomes without heterogeneity. Relative risks for those outcomes with significant heterogeneity were calculated with the random-effects model. The results of both models were reported for each outcome that was assessed. The null hypothesis underlying the overall test of association was that the overall relative risk was equal to one.

Publication bias was examined by reviewing the Egger test and by visual inspection of funnel plots, which plot relative risk against study sample size.[26] When biases and heterogeneity were not present, the variation in the estimated effect decreases with increasing sample size, and the plot resembles a symmetric funnel. Asymmetry in such funnel plots is usually caused by small trials reporting greater effects, on average, than large trials, suggesting publication or other biases.

We performed sensitivity analyses to establish the robustness of our results. To identify any study that may have exerted a disproportionate influence on the summary treatment effect, we deleted studies one at a time and noted the degree to which the size and significance of treatment effect changed. No study exerted a majority of influence in the meta-analysis for either of the outcomes assessed.

To determine the combinability of individual studies, we performed a formal test of homogeneity treatment effect across studies using the Breslow–Day method.[27] In addition, homogeneity across studies was assessed by qualitative visual inspection of L'Abbé plots.[28] We also performed several preplanned sensitivity analyses to assess the influence of individual studies. In sensitivity analysis, the meta-analysis estimates are computed omitting one study at a time and noting the degree to which the size and significance of the treatment effect change.

To explore other potential sources of study result heterogeneity, we developed a random effects regression model that included several variables as covariates. In this meta-regression, the dependent variable was the relative risk for the outcome of interest (malignancy) and the independent variables were total number of patients, year, country of study, and tamoxifen use.

A cumulative meta-analysis was also executed. This statistical technique allows researchers to measure how much the observed findings change as evidence accumulates so that the effect of adding data from each study on the overall findings can be determined. In this cumulative meta-analysis, studies

Table 1. Study Characteristics

Reference (Lead Author)	Year	Country	Patients (n)	Patients With Malignant Polyps (n)
Anasasiadis[43]	1999	Greece	126	11
Orvieto[33]	1999	Israel	146	4
Goldstein[19]	2002	United States	61	7
Ben-Arie[14]	2003	Israel	420	27
Savelli[31]	2003	Itlay	509	20
Shushan[30]	2004	Israel	300	4
Martinez[34]	2004	Spain	1,822	16
Fernandez-Parra[38]	2006	Spain	653	10
Antunes[42]	2006	Brazil	455	18
Lieng[35]	2007	Norway	411	14
Dreisler[39]	2008	Denmark	48	3
Rahimi[32]	2009	Italy	694	54
Baiocchi[41]	2009	Italy	1,242	60
Domingues[40]	2009	Portugal	481	22
Ferrazzi[37]	2009	Italy	1,922	64
Wang[29]	2009	China	766	29
Gregoriou[36]	2009	Greece	516	14
Total			10,572	377

were ordered by control event rate (the prevalence of malignancy in premenopausal women found to have endometrial polyps and the prevalence of malignancy in women free of abnormal bleeding found to have endometrial polyps).

RESULTS

We identified 1,552 publications evaluating the risk of malignancy within endometrial polyps. The study flow of these publications is shown in Figure 1. Of the 1,552 studies originally identified, 1,295 were excluded as a result of lack of suitable study design. Additionally, 225 studies were excluded because our inclusion criteria were not met. Fifteen articles were excluded because neither menopausal status nor bleeding status was reported. Seventeen studies were ultimately included[14,19,29–43]; 13 of these reported on menopausal status,[14,29–32,34–36,38,39,41–43] and 13 reported on bleeding status.[14,19,29–31,33–35,37–41]

A summary of these observational studies is shown in Table 1. A total of 10,572 patients underwent polypectomy with histopathologic analysis in these 17 studies. Three hundred seventy-seven polyps were malignant (3.57%). Four studies were conducted in Italy, three in Israel, two in Spain, two in Greece, one in Denmark, one in Norway, one in Portugal, one in Brazil, one in China, and one in the United States. The studies differed in study design, size, inclusion criteria, and exclusion criteria.

With respect to menopausal status, endometrial neoplasia was identified in 5.42% (214 of 3,946) of women with endometrial polyps who were postmenopausal (relative risk 3.86; 95% CI 2.92–5.11) compared with 1.70% (68 of 3,997) of premenopausal women (Table 2). There was no evidence of significant heterogeneity noted for this outcome variable (P=.25).

With respect to abnormal uterine bleeding, 4.15% (195 of 4,697) of women with symptomatic bleeding had neoplastic polyps (relative risk 1.97; 95% CI 1.24–3.14) compared with 2.16% (85 of 3,941) of women without abnormal bleeding (Table 3). There was evidence of significant heterogeneity noted for this outcome variable (P=.006); accordingly, the random-effects model results were reported. Funnel plots did not identify evidence of publication bias.

The presence of endometrial neoplasia among postmenopausal women with bleeding was specifically assessed. Among postmenopausal women with bleeding and endometrial polyps, 4.47% (88 of 1,968) had a malignant polyp in comparison to 1.51% (25 of 1,654) of asymptomatic postmenopausal women (relative risk 3.36; 95% CI 1.45–7.80). Statistically significant heterogeneity was noted for this outcome variable (P=.005) and the random-effects model was reported. There was no evidence of publication bias noted on the funnel plots.

Eight studies assessed the relationship of polyp size with the risk of malignancy.[14,19,29,30,32,36–38] Four studies suggested that larger polyps are associated with a higher risk of polyp malignancy.[14,29,32,37] Fernandez-Parra et al,[38] Goldstein et al,[19] Shushan et al,[30] and Gregoriou et al[36] stated that polyp size is not a distinctive feature of malignancy. The authors reported the size of polyps in

Table 2. Menopausal Status and Risk of Malignancy

Reference (Lead Author)	Postmenopausal	Premenopausal	Relative Risk	95% Confidence Interval
Dreisler[39]	3/24 (12.5)	0/24 (0)	7.00	0.38–128.61
Fernandez-Parra[38]	10/375 (2.67)	0/277 (0)	15.28	0.91–263.84
Rahimi[32]	32/241 (13.28)	22/453 (4.86)	2.73	1.62–4.60
Ben-Arie[14]	22/218 (10.09)	5/184 (2.72)	3.71	1.43–9.61
Savelli[31]	18/317 (5.68)	2/192 (1.04)	5.45	1.28–23.35
Baiocchi[41]	49/585 (8.38)	11/657 (1.67)	5.00	2.62–9.53
Anasasiadis[43]	11/75 (14.67)	0/51 (0)	15.74	0.95–261.22
Shushan[30]	2/90 (2.22)	2/210 (0.95)	2.33	0.334–16.31
Lieng[35]	9/204 (4.41)	5/207 (2.42)	1.83	0.62–5.36
Antunes[42]	14/326 (4.29)	4/129 (3.10)	1.39	0.47–4.13
Martinez[34]	14/994 (1.41)	2/828 (0.24)	5.83	1.33–25.58
Wang[29]	16/104 (15.38)	13/662 (1.96)	7.83	3.88–15.81
Gregoriou[36]	14/393 (3.56)	2/123 (1.63)	2.19	0.50–9.51
Total	214/3,946 (5.42)	68/3,997 (1.70)		
M-H pooled RR			3.86	2.93–5.11

M-H, Mantel-Haenszel; RR, relative risk.
Data are n/N (%) unless otherwise specified.
Heterogeneity χ^2=14.86 (*P*=.249). I^2 (variation in RR attributable to heterogeneity)=19.3%. Test of RR=1: z=9.51, *P*<.01.

centimeters, millimeters, grams, and milliliters. The data regarding polyp size and risk of malignancy were not amenable to meta-analysis because of the differing units of measurement.

Our cumulative meta-analysis indicated that even limiting analysis to the two studies with the lowest prevalence of malignancy among premenopausal women with endometrial polyps, the elevated risk of malignancy among menopausal women with polyps achieved statistical significance. Likewise, even limiting analysis to the three studies with the lowest prevalence of malignancy among women free of abnormal bleeding found to have endometrial polyps, the elevated risk of malignancy among women with abnormal bleeding and endometrial polyps achieved statistical significance (Table 4).

Meta-regression analysis with total number of patients, year, country of study, and tamoxifen use as covariates did not alter results (Table 5).

DISCUSSION

This systematic review demonstrates that among women found to have endometrial polyps, symptomatic vaginal bleeding and postmenopausal status are

Table 3. Bleeding Status and Risk of Malignancy

Reference (Lead Author)	Symptomatic Uterine Bleeding	Asymptomatic	Relative Risk	95% Confidence Interval
Dreisler[39]	1/9 (11.11)	2/39 (5.13)	2.17	0.22–21.36
Domingues[40]	21/181 (11.60)	1/175 (0.57)	30.30	2.76–149.32
Fernandez-Parra[38]	10/470 (2.13)	0/183 (0)	8.20	0.48–139.28
Ferrazzi[37]	46/1,152 (3.99)	18/770 (2.34)	1.71	1.00–2.92
Orvieto[33]	3/51 (5.88)	1/95 (1.05)	5.59	0.60–52.36
Goldstein[19]	4/42 (9.52)	3/19 (15.79)	0.60	0.15–2.44
Ben-Arie[14]	12/188 (6.38)	15/214 (7.01)	0.91	0.44–1.90
Savelli[31]	11/274 (4.01)	9/235 (3.83)	1.05	0.44–2.49
Baiocchi[41]	39/397 (9.82)	21/845 (2.49)	3.96	2.36–6.63
Shushan[30]	4/227 (1.76)	0/73 (0)	2.92	0.16–53.62
Lieng[35]	9/282 (3.19)	5/129 (3.88)	0.82	0.28–2.41
Martinez[34]	12/998 (1.20)	4/824 (0.49)	2.48	0.80–7.65
Wang[29]	23/426 (5.40)	6/340 (1.76)	3.06	1.26–7.43
Total	195/4,697 (11.11)	85/3,941 (5.13)		
D+L pooled RR			1.976	1.245–3.137

D+L, DerSimonian and Laird; RR, relative risk.
Data are n/N (%).
Heterogeneity χ^2=27.58 (*P*=.006). I^2 (variation in RR attributable to heterogeneity)=56.5%. Test of RR=1: z=2.89, *P*=.004.

Table 4. Cumulative Meta-Analysis for Control Event Rate (Postmenopausal Women)

Reference (Lead Author)	Cumulative Estimate	95% Confidence Interval	z	P
Dreisler[39]	15.53	0.91–263.84	1.90	.058
Fernandez-Parra[38]	15.63	2.13–114.90	2.70	.007
Rahimi[32]	12.09	2.33–62.68	2.97	.003
Ben-Arie[14]	8.08	2.69–24.26	3.72	0
Savelli[31]	5.98	2.30–15.57	6.66	0
Baiocchi[41]	5.81	2.62–12.92	4.32	0
Anasasiadis[43]	4.65	2.31–9.38	4.29	0
Shushan[30]	4.84	3.01–7.77	6.51	0
Lieng[35]	5.63	3.80–8.34	8.61	0
Antunes[42]	4.93	3.41–7.13	8.47	0
Martinez[34]	4.75	3.37–6.70	8.87	0
Wang[29]	4.25	3.06–5.90	8.64	0
Gregoriou[36]	3.75	2.84–4.95	9.33	0

associated with an increased risk of endometrial neoplasia. It also suggests that polyp size does not independently predict risk of malignancy. The fact that menopausal status and abnormal bleeding were found to be significant predictors of polyp malignancy even in those populations with the lowest risk of malignancy (among premenopausal women or those without abnormal bleeding) adds to our confidence that these two parameters represent important risk factors for malignancy in women with polyps.

We included endometrial hyperplasia with atypia in the malignancy group for two related reasons. First, among women with atypical endometrial hyperplasia on endometrial biopsy, the rate of concurrent endometrial carcinoma diagnosed at the time of hysterectomy is 42.6%.[21] In addition, approximately 28% of women with untreated complex hyperplasia with atypia progress to carcinoma.[44] The overall risk of malignancy among all women with endometrial polyps in this meta-analysis was 377 of 10,572 (3.57%). This systematic review and meta-analysis provides us with a more precise estimate of malignancy in women with endometrial polyps than is currently available.

Although systematic reviews with meta-analyses provide an explicit method for synthesizing evidence and demonstrating early evidence with regard to the effectiveness of treatments, they may not be as valuable as a single large observational study. In this review, no single study accounted for the majority of the patients analyzed. Of the 17 studies reviewed, the findings of 12[14,29,31,32,34–36,38,39,41–43] revealed an increased risk of malignancy in women with postmenopausal status. Similarly, of the 13 reports assessing this outcome, the findings of six[29,30,37,38,40,41] revealed an increased risk of malignancy in women with abnormal bleeding patterns.

We did not limit our search to publications published in English. No difference in the quality of studies published in English compared with other languages has been demonstrated.[46] Authors are more likely to publish in an international, English-language journal if results are positive, whereas negative findings are more often published in a local journal.[46,47] Limiting meta-analysis to publications in English-language journals can therefore restrict the completeness of information retrieval, thereby causing bias. One of the studies included in our review was published in Spanish.[34]

An important issue related to meta-analysis is heterogeneity. In performing meta-analysis, one assumes similarity among the various evaluated studies. Although absolute homogeneity among studies might appear desirable, it could in fact preclude generalization to a broad clinical population; nevertheless, understanding the sources of heterogeneity remains important.

We found significant heterogeneity for the abnormal uterine bleeding variable evaluated by this review. There are several possible explanations for the observed heterogeneity among studies. Fourteen studies included were retrospective,[14,29–38,40–42] whereas three were prospective observational studies.[19,39,43] The diagnosis of endometrial polyps was made by three different methods: transvaginal (unenhanced) ultrasound, saline infusion sonohysterography, and hysteroscopy. Some studies used a combination of these techniques.

In three studies, only women with postmenopausal bleeding were assessed.[33,37,40] Fourteen studies included

Table 5. Meta-Regression Analysis

	Coefficient	Standard Error	Z-Score	P>Z	95% Confidence Interval
Year	0.115	0.081	1.42	.155	−0.044 to 0.274
Country	0.612	0.549	1.11	.265	−0.464 to 1.688
Tamoxifen	0.011	0.462	0.02	.980	−0.895 to 0.918
Subjects	−0.000	0.000	−0.58	.560	−0.001 to 0.001

premenopausal and postmenopausal patients with abnormal bleeding patterns.[14,19,29–32,34–36,38,39,41–43]

Ten studies included women using tamoxifen,[14,29–31,34,35,38,41,42,47] nine included women using menopausal therapy,[14,29,31,33,34,36,38,41,42] and two included women using oral contraceptives.[31,36] Many studies have demonstrated that tamoxifen use increases the risk of endometrial cancer and the risk of malignancy in endometrial polyps.[48–58] Martinez[34] was the only author in this series who reported a significantly elevated risk of malignancy in endometrial polyps with the use of tamoxifen. With regard to menopausal hormone therapy, Orvieto[33] was the sole author to find a significant association with malignant polyps. None of the studies in this meta-analysis found a relationship between oral contraceptive use and malignancy within endometrial polyps.

Obesity was assessed as an independent risk factor of malignancy within the polyps in four studies.[29,35,36,42] Women, who are obese, have higher levels of circulating estrogen, which stimulate the endometrium to produce endometrial polyps and possibly malignant endometrial polyps.[59–61] Only one study included in this meta-analysis found a statistically significant correlation between obesity and malignant endometrial polyps.[36]

The correlation between diabetes mellitus and malignancy within endometrial polyps has been explored by several authors.[29,31,33,36,42] One author reported a statistically significant association between diabetes and polyp malignancy.[36] Four authors found no relationship between the two disease processes.[29,31,33,42]

Similarly, hypertension was examined as an independent risk factor of malignancy within endometrial polyps by five authors who were included in this meta-analysis.[29,31,33,36,41] Savelli and Baiocchi were the only authors who reported a statistically significant association between these two disorders.[31,41]

Based on our systematic review and meta-analysis, we conclude that among women with endometrial polyps, the presence of abnormal bleeding or menopausal status is associated with an increased risk of endometrial neoplasia. These findings should inform the counseling and care of women with endometrial polyps.

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